Class Imbalance and Domain Shift in Vehicle-Dynamics-Based Drowsy Driving Detection

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Abstract—Vehicle-dynamics-based drowsy driving detection (DDD) is a non-intrusive and practical approach to driver monitoring, but its performance may be degraded by two issues: class imbalance, because drowsy samples are rare, and domain shift, because driving behavior varies across drivers. In vehicle-dynamics-based DDD, however, it remains unclear whether these issues materially affect detection performance and which countermeasures are effective. To address this question, we construct a DDD framework using a public 87-driver simulator dataset and conduct a four-factor factorial experiment varying distance metric (D), domain membership (G), training mode (M), and rebalancing strategy (R). Sobol–Hoeffding variance decomposition shows that training mode and rebalancing account for more than 95% of systematic variance, with a substantial $R \times M$ interaction that reverses the optimal rebalancing strategy across training modes. In contrast, distance metric and domain membership have no measurable effect on performance. A subject-wise application of SMOTE (SW-SMOTE) at ratio $r = 0.1$ with within-domain training maximizes threshold-independent discrimination (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558). These results establish a clear design priority for vehicle-dynamics-based DDD: class rebalancing and training mode selection are the dominant performance levers, whereas domain grouping configuration can be safely omitted from the optimization process.

Index Terms—Drowsy driving detection, class imbalance, domain shift, vehicle dynamics, Sobol sensitivity analysis, factorial experiment, SMOTE.

I. Introduction

DROWSY driving is a leading cause of traffic accidents worldwide, accounting for an estimated 15–20% of road fatalities [1]. Vehicle-dynamics-based drowsy driving detection (DDD) offers a non-intrusive and practical approach by monitoring signals such as steering angle, lateral acceleration, and lane offset without requiring sensors attached to the driver [2], [3].

Two challenges are potentially critical for vehicle-dynamics-based DDD. First, class imbalance: drowsy episodes are inherently rare in driving data, with alert states far outnumbering drowsy states, which may bias classifiers toward the majority class and impair detection of safety-critical events. Second, domain shift: driving behavior varies across individuals. Because training data typically cover only a subset of potential drivers (in-domain), a deployed model must generalize to unseen drivers (out-domain), whose driving patterns may differ substantially. This inter-driver variability may degrade detection performance.

Improving detection performance under these conditions is a practical priority, yet the degree to which each factor actually degrades performance has not been quantified. Although class imbalance handling [4], [5] and domain adaptation [6], [7] have been studied extensively in other contexts, their importance in vehicle-dynamics-based DDD remains unclear. In particular, it is not known whether these issues materially degrade performance in this modality, whether they warrant dedicated countermeasures, and how they interact when both are present. To this end, we pose the following research questions:

- RQ1. To what extent do class imbalance and domain shift each affect vehicle-dynamics-based DDD performance, as quantified through the four design factors (D, G, M, R)?
- RQ2. Do the contributing factors interact, and if so, what is the optimal factor combination?

This paper addresses these research questions by constructing a DDD framework using a public open dataset and conducting a four-factor factorial experiment to quantify the effects of class imbalance handling and domain configuration on detection performance. The four factors—distance metric (D), domain membership (G), training mode (M), and rebalancing strategy (R)—are systematically varied across 1,512 random forest runs, and Sobol–Hoeffding variance decomposition is applied to identify which factors significantly affect performance and which combinations are practically effective. Of these, distance metric (D), domain membership (G), and training mode (M) address domain configuration, while rebalancing strategy (R) addresses class imbalance. The main contributions are:

1) We construct a DDD experimental framework on a publicly available 87-driver dataset that enables joint, controlled analysis of class imbalance handling and domain shift under a unified factorial design.
2) Through a four-factor factorial experiment with Sobol–Hoeffding variance decomposition, we quantify the relative importance of these design factors and reveal that training mode and class rebalancing dominate (> 95% of systematic variance), with a substantial $R \times M$ interaction that reverses the optimal strategy across training modes.

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3) We empirically position subject-wise SMOTE (SW-SMOTE)—an instance of cluster-based SMOTE [8] with clusters defined by driver identity, motivated by the subject-dependency concerns identified for physiological-signal augmentation [9]—within the four-factor design, and show through exhaustive factor-combination search that SW-SMOTE with within-domain training maximizes threshold-independent discrimination (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558).

The remainder of this paper is organized as follows. Section II reviews related work. Section III formalizes the DDD task and the two data challenges (class imbalance and domain shift). Section IV describes the four-factor experimental design, common settings, and analysis methods. Section V reports experimental results from OFAT, Sobol decomposition, interaction analysis, and robustness checks. Section VI answers the research questions and discusses their practical implications. Section VII concludes with a summary and future directions.

II. Related Work

A. Vehicle-Dynamics-Based Drowsy Driving Detection

Vehicle dynamics signals—steering wheel angle, lateral acceleration, and lane position—are widely used for DDD because they enable non-intrusive monitoring. Arefnezhad et al. [10] used statistical and spectral features with SVM, Baccour et al. [11] compared vehicle-based and driver-based features for drowsiness monitoring using SVM with class weighting, and Sun et al. [12] optimized per-driver calculation parameters to account for driver fingerprinting differences. Zhao et al. [13] applied continuous wavelet transform features to steering signals with SVM, and Wang et al. [14] used LSTM on five vehicle signals, although their study targeted driver distraction rather than drowsiness. More recently, Zhu et al. [15] proposed MADFM, a multi-layer adaptive fatigue monitoring model based on steering wheel signals. Recent surveys [2], [3], [16], [17] confirm the practical relevance of this modality. However, most studies evaluate a single method on a fixed split, and the sensitivity of detection performance to data preprocessing choices has received little attention.

B. Class Imbalance in Drowsy Driving Detection

Class imbalance is inherent in drowsy driving data because drowsy episodes are rare but safety-critical. He and Garcia [5] summarize major countermeasures such as random under-sampling (RUS), Synthetic Minority Over-sampling Technique (SMOTE) [4], and cost-sensitive learning, but none of these standard techniques account for the multi-subject structure of pooled driving data. In vehicle-dynamics-based DDD, only a few studies explicitly mention imbalance handling: Yadawadkar et al. [18] compared SMOTE with no resampling, Schwarz et al. [19] used SMOTE in steering-based detection, and Baccour et al. [11] used class weighting when comparing vehicle-dynamics-based and driver-based features. However, these studies treat imbalance handling as an auxiliary choice rather than the main object of study, and systematic comparison of multiple countermeasures remains lacking. Moreover, the effectiveness of resampling may depend on the characteristics of the target class, such as temporal clustering and inter-subject variability in drowsy episodes, which require dedicated treatment beyond standard techniques.

C. Domain Shift in Drowsy Driving Detection

Inter-subject variability creates domain shift in DDD [6], [20]. In vehicle-dynamics-based DDD and related driver-state studies, Sun et al. [12] examined subject-independent drowsiness detection while highlighting driver fingerprinting differences, and Maritsch et al. [21] used leave-one-subject-out evaluation for vehicle-dynamics-based driver-state prediction. These studies recognize cross-subject variability, but systematic comparison of domain-related design choices in vehicle-dynamics-based DDD remains limited.

III. Problem Formulation

A. DDD Framework Overview

Vehicle-dynamics-based DDD monitors vehicle signals—such as steering angle, lateral acceleration, and lane position—to infer the driver's alertness level without requiring on-body sensors. Fig. 1 sketches the generic processing pipeline that is shared, with minor variations, across most prior vehicle-dynamics-based DDD studies: per-subject signal segmentation and labeling, feature extraction, chronological train/validation/test split, classifier training, probability calibration, threshold optimization, and held-out test-set evaluation.

Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ denote the set of N subjects. For each subject s, vehicle signals are segmented into fixed-length time windows, and each window is assigned a binary drowsiness label based on a subjective sleepiness scale, yielding a per-subject dataset $\mathcal{D}_s = \{(\mathbf{x}_w, y_w)\}_{w=1}^{n_s}$, where $\mathbf{x}_w \in \mathbb{R}^p$ is a feature vector extracted from the window and $y_w \in \{0, 1\}$ is the drowsiness label (0: alert, 1: drowsy).

Each subject's dataset is split chronologically into three disjoint portions:

$$\mathcal{D}_s = \mathcal{D}_s^{\mathrm{tr}} \mathbin{\dot\cup} \mathcal{D}_s^{\mathrm{val}} \mathbin{\dot\cup} \mathcal{D}_s^{\mathrm{te}} \tag{1}$$

with a 70/15/15 ratio, preserving temporal order to prevent data leakage from future windows. The training set $\mathcal{D}_s^{\mathrm{tr}}$ is used to fit the classifier (with hyperparameter selection performed via cross-validation within $\mathcal{D}_s^{\mathrm{tr}}$), the validation set $\mathcal{D}_s^{\mathrm{val}}$ for threshold tuning, and the test set $\mathcal{D}_s^{\mathrm{te}}$ for final performance evaluation.

The DDD task is to learn a scoring function $f\colon \mathbb{R}^p \to [0, 1]$ that generalizes to held-out test windows, trained on pooled data $\mathcal{D}_{\mathrm{pool}} = \bigcup_{s \in \mathcal{S}_M} \mathcal{D}_s^{\mathrm{tr}}$, where $\mathcal{S}_M \subseteq \mathcal{S}$ is the training subject set determined by mode M (Eq. (7)); the binary prediction is obtained by thresholding: $\hat{y}_w = \mathbf{1}[f(\mathbf{x}_w) \geq \hat{\tau}]$, where $\hat{\tau}$ is the F2-optimal operating point (Section ??). The specific dataset, signal set, and preprocessing parameters are detailed in Section IV-C.

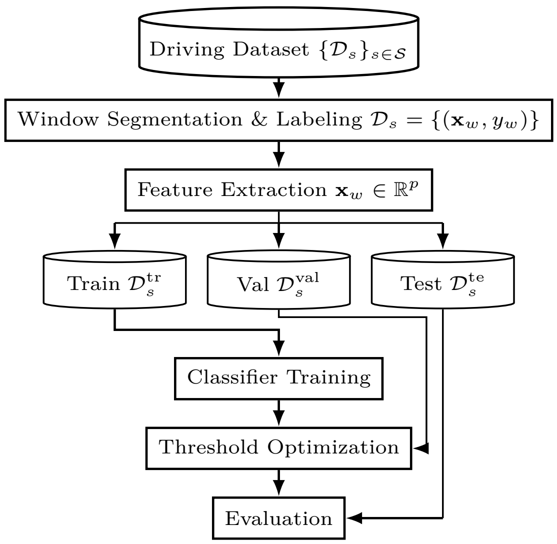

Fig. 1. Generic vehicle-dynamics-based DDD pipeline. Per-subject vehicle signals are segmented into labeled windows, features are extracted, and a chronological train/validation/test split is applied. The training set fits the classifier (which absorbs hyperparameter tuning and probability calibration as standard sub-steps); the validation set tunes the decision threshold; the test set is reserved for the final evaluation. This pipeline is shared, with minor variations, across most prior DDD studies; the four experimental factors specific to this work are introduced in Section IV (Fig. 2).

B. Class Imbalance

For a given dataset $\mathcal{D}$, let $n_c(\mathcal{D}) = |\{(\mathbf{x}_w, y_w) \in \mathcal{D} : y_w = c\}|$ denote the number of windows with label c, where $c = 0$ (alert) and $c = 1$ (drowsy). Because drowsy episodes are rare, $n_1 \ll n_0$, yielding an imbalance ratio $\rho = n_1/n_0 \ll 1$. A rebalancing operator transforms the training set $\mathcal{D}$ into $\tilde{\mathcal{D}}$ so that the adjusted ratio $\tilde{n}_1/\tilde{n}_0$ approaches a target value r, either by under-sampling the majority class or over-sampling the minority class.

C. Domain Shift

Each subject s induces a marginal feature distribution $P_s(\mathbf{x})$. Inter-driver variability means that $P_{s_i}(\mathbf{x}) \neq P_{s_j}(\mathbf{x})$ in general; following the transfer-learning taxonomy of Pan and Yang [6], this constitutes covariate shift when the labeling mechanism $P(y \mid \mathbf{x})$ is approximately shared across subjects. When subjects are partitioned into a source group $\mathcal{G}_{\mathrm{src}}$ and a target group $\mathcal{G}_{\mathrm{tgt}}$ with $\mathcal{G}_{\mathrm{src}} \cap \mathcal{G}_{\mathrm{tgt}} = \emptyset$, a classifier trained on $\bigcup_{s\in\mathcal{G}_{\mathrm{src}}} \mathcal{D}_s$ may degrade on $\bigcup_{s\in\mathcal{G}_{\mathrm{tgt}}} \mathcal{D}_s$ because the pooled training distribution differs from the target distribution.

Both class imbalance (Section III-B) and domain shift (Section III-C) are treated here not as fixed nuisances but as explicit design factors whose practical importance is quantified through the factorial experiment in Section IV.

IV. Experimental Framework

To answer the two research questions posed in Section I—which factors materially affect performance (RQ1) and whether their interactions determine the optimal design (RQ2)—we design a four-factor factorial experiment. Sobol–Hoeffding variance decomposition (Section IV-D) quantifies each factor's individual and interactive contribution, enabling a principled answer to both questions from a single unified experiment.

Section IV-A provides an overview of the factorial design; Section IV-B defines each factor; Section IV-C describes common experimental settings; and Section IV-D presents the analysis methods.

A. Overview of the Factorial Experiment

The experiment follows a four-factor full-factorial design. Fig. 2 shows the dependency chain $D \to G \to M \to R$ of the four study-specific factors, where each factor acts on the data partition constructed by its predecessors; the resulting (R, M, D, G) cell is then evaluated using the generic DDD pipeline of Fig. 1. This chain operationalizes the formulation of Section III as follows.

Starting from the subject set $\mathcal{S}$ with per-subject datasets $\mathcal{D}_s = \{(\mathbf{x}_w, y_w)\}$ (Section III-A), a distance metric D first computes pairwise inter-subject distances and partitions $\mathcal{S}$ into an in-domain group $\mathcal{G}_{\mathrm{in}}$ (44 subjects, most typical) and an out-domain group $\mathcal{G}_{\mathrm{out}}$ (43 subjects, most dissimilar). Domain membership G then selects which of these two groups serves as the evaluation target $\mathcal{G}_{\mathrm{eval}}$. Each subject's data is split chronologically into $(\mathcal{D}_s^{\mathrm{tr}}, \mathcal{D}_s^{\mathrm{val}}, \mathcal{D}_s^{\mathrm{te}})$ per Eq. (1) (70/15/15 ratio). Training mode M assembles the pooled training set $\mathcal{D}_{\mathrm{pool}} = \bigcup_{s\in\mathcal{S}_M} \mathcal{D}_s^{\mathrm{tr}}$ from a subset $\mathcal{S}_M \subseteq \mathcal{S}$ determined by the partition. After feature extraction and selection ($p = 90 \to k = 10$ Arefnezhad descriptors per Section IV-C2), rebalancing strategy R applies the operator of Section III-B to produce $\tilde{\mathcal{D}}$ with adjusted class ratio approaching the target r. The classifier $f : \mathbb{R}^p \to \{0, 1\}$ (Section III-A) is trained on $\tilde{\mathcal{D}}$, calibrated, and evaluated on $\mathcal{D}_{\mathrm{test}} = \bigcup_{s\in\mathcal{G}_{\mathrm{eval}}} \mathcal{D}_s^{\mathrm{te}}$.

The four factors thus form a dependency chain $D \to G \to M \to R$, where each downstream factor operates on the data subset constructed by its predecessors. Each factorial cell is uniquely determined by a quadruple (R, M, D, G); Section IV-B defines the levels of each factor (summarized in Table I) and the total number of observations.

B. Factor Definitions

Table I summarizes the four factors and their levels. The $7 \times 3 \times 3 \times 2 = 126$ factor combinations are each evaluated over 12 fixed random seeds, yielding $N = 1{,}512$ total observations. The following subsubsections define each factor in detail.

1) Distance Metric (D) and Domain Grouping: To define the binary domain partition $(\mathcal{G}_{\mathrm{in}}, \mathcal{G}_{\mathrm{out}})$ on which factors G and M operate, pairwise distances $d(s_i, s_j)$ between all 87 subjects are computed from the per-subject window sets $\mathcal{D}_{s_i}$ and $\mathcal{D}_{s_j}$ using three metrics:

TABLE I
Factorial Experiment Design

Factor	Sym.	Lvl.	Values
Rebalancing	R	7	BL, RUS/SMOTE/SW-SMOTE $r \in \{0.1, 0.5\}$
Training mode	M	3	Cross, Within, Mixed
Distance	D	3	MMD, DTW, Wasserstein
Membership	G	2	In-domain, Out-domain

Maximum Mean Discrepancy (MMD) [22] measures distributional difference in a reproducing kernel Hilbert space $\mathcal{H}$:

$$\mathrm{MMD}^2(s_i, s_j) = \left\| \frac{1}{n_{s_i}} \sum_{w \in s_i} \phi(\mathbf{x}_w) - \frac{1}{n_{s_j}} \sum_{w \in s_j} \phi(\mathbf{x}_w) \right\|_{\mathcal{H}}^2 \tag{2}$$

where ϕ is the feature map induced by a Gaussian kernel.

Dynamic Time Warping (DTW) [23] aligns two univariate time series and finds the warping path that minimizes cumulative Euclidean distance. Each subject's multivariate window sequence is first reduced to a scalar series $\bar{x}_t^{(i)} = \frac{1}{q} \sum_{d=1}^{q} x_{t,d}^{(i)}$ (mean across all $q = 145$ z-scored features at each time step), and DTW is computed as:

$$\mathrm{DTW}(s_i, s_j) = \sqrt{\min_{\pi \in \mathcal{P}} \sum_{(a,b) \in \pi} \left(\bar{x}_a^{(i)} - \bar{x}_b^{(j)} \right)^2} \tag{3}$$

where $\mathcal{P}$ is the set of admissible warping paths. Because DTW with Euclidean cost is symmetric, no explicit symmetrization is required.

Wasserstein Distance [24] computes the minimum transport cost between the empirical feature distributions:

$$W_1(s_i, s_j) = \inf_{\gamma \in \Gamma(P_{s_i}, P_{s_j})} \int \|\mathbf{x} - \mathbf{x}'\| \, d\gamma(\mathbf{x}, \mathbf{x}') \tag{4}$$

where $\Gamma(P_{s_i}, P_{s_j})$ is the set of couplings with marginals P_{s_i} and P_{s_j}.

For each metric, a K-nearest-neighbor (KNN) score ($K = 5$) is computed per subject as the mean distance to its K nearest neighbors:

$$\mathrm{KNN}_K(s) = \frac{1}{K} \sum_{j \in \mathcal{N}_K(s)} d(s, s_j) \tag{5}$$

Subjects are split at the median of KNN_K into in-domain (44 subjects, lowest scores, most typical) and out-domain (43 subjects, highest scores, most dissimilar), forming two groups $\mathcal{G}_{\text{in}}$ (in-domain, 44 subjects) and $\mathcal{G}_{\text{out}}$ (out-domain, 43 subjects); their roles as evaluation target or training source are determined subsequently by G and M (Eqs. (6)–(7)).

Distance metric $D \in \{\mathrm{MMD}, \mathrm{DTW}, W_1\}$ determines the pairwise subject distances and the binary typicality-based partition $(\mathcal{G}_{\text{in}}, \mathcal{G}_{\text{out}})$ via Eqs. (2)–(5).

TABLE II
Vehicle Dynamics Signals

Symbol	Signal	Unit
$\delta(t)$	Steering angle	rad
$\dot{\delta}(t)$	Steering speed	rad/s
$a_y(t)$	Lateral acceleration	m/s^2
$a_x(t)$	Longitudinal acceleration	m/s^2
$e_{\text{lane}}(t)$	Lane offset	m

2) Domain Membership (G): Domain membership $G \in \{\text{in}, \text{out}\}$ selects the evaluation-target group:

$$\mathcal{G}_{\text{eval}} = \begin{cases} \mathcal{G}_{\text{in}} & \text{if } G = \text{in (44 subjects)} \\ \mathcal{G}_{\text{out}} & \text{if } G = \text{out (43 subjects)} \end{cases} \tag{6}$$

3) Training Mode (M): Training mode $M \in \{\text{Cross}, \text{Within}, \text{Mixed}\}$ assembles the training pool from subjects determined by $\mathcal{G}_{\text{eval}}$:

$$\mathcal{D}_{\text{pool}} = \bigcup_{s \in \mathcal{S}_M} \mathcal{D}_s^{\text{tr}}, \quad \mathcal{S}_M = \begin{cases} \mathcal{S} \setminus \mathcal{G}_{\text{eval}} & M = \text{Cross} \\ \mathcal{G}_{\text{eval}} & M = \text{Within} \\ \mathcal{S} & M = \text{Mixed} \end{cases} \tag{7}$$

Cross-domain trains exclusively on the opposite group, within-domain on the evaluation group itself, and mixed on all 87 subjects.

4) Rebalancing Strategy (R): Rebalancing strategy R applies a rebalancing operator $\mathcal{R}$ to the training pool, yielding $\tilde{\mathcal{D}} = \mathcal{R}(\mathcal{D}_{\text{pool}})$ with adjusted class ratio $\tilde{n}_1/\tilde{n}_0$. Seven levels are defined:

- Baseline (BL): $\tilde{\mathcal{D}} = \mathcal{D}_{\text{pool}}$ (identity; original ratio ρ preserved).
- RUS at $r \in \{0.1, 0.5\}$: randomly discard majority-class samples until $\tilde{n}_1/\tilde{n}_0 = r$.
- SMOTE [4] at $r \in \{0.1, 0.5\}$: generate synthetic minority-class samples via k-NN interpolation on $\mathcal{D}_{\text{pool}}$ until $\tilde{n}_1/\tilde{n}_0 = r$.
- SW-SMOTE at $r \in \{0.1, 0.5\}$: apply SMOTE independently within each subject before pooling, an instance of cluster-based SMOTE [8] with clusters fixed to subject identity:

$$\tilde{\mathcal{D}} = \bigcup_{s \in \mathcal{S}_M} \mathrm{SMOTE}_r(\mathcal{D}_s^{\text{tr}}) \tag{8}$$

so that synthetic samples preserve individual driving patterns rather than interpolating across subjects.

The classifier f is trained on $\tilde{\mathcal{D}}$ and evaluated on $\mathcal{D}_{\text{test}} = \bigcup_{s \in \mathcal{G}_{\text{eval}}} \mathcal{D}_s^{\text{te}}$.

C. Common Experimental Settings

1) Dataset and Signals: The publicly available driving dataset of Aygun et al. [25] is used, comprising $N = 87$ subjects each recorded over two sessions on the SIMlsl driving simulator at $f_s = 60$ Hz. Five raw signals are extracted (Table II).

Under the kinematic bicycle model [26], lateral acceleration follows:

$$a_y = \frac{v^2}{L} \delta \tag{9}$$

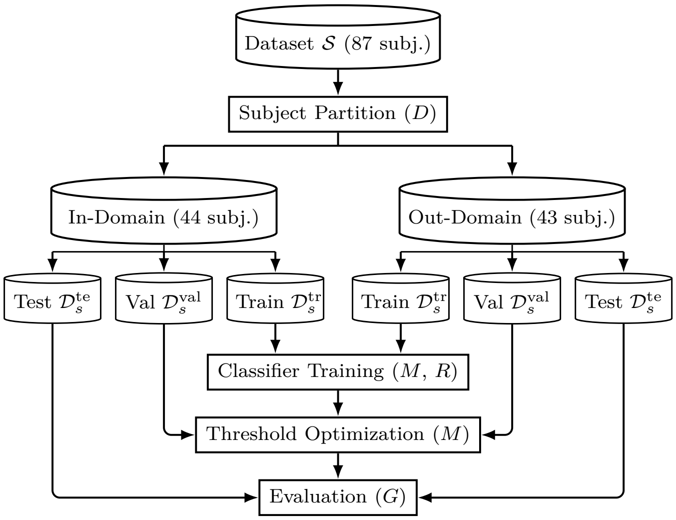

Fig. 2. Factorial experimental framework. Activity blocks are plain white: the upper half (Subject Partition → In/Out groups → per-subject 70/15/15 split) reuses the data-partition stage of the original combined-framework figure; the lower half mirrors the three terminal blocks of Fig. 1 (Classifier Training → Threshold Optimization → Evaluation), absorbing training-pool assembly, feature selection, rebalancing, hyperparameter tuning, and probability calibration into the Classifier Training block. Following the dependency chain $D \to G \to M \to R$, each factor is annotated inline at the activity it configures: Subject Partition by D (Distance Metric); Evaluation by G (Domain Membership); Classifier Training by M (Training Mode — selects the subject group whose Train splits form the training pool) and R (Rebalancing — applied to the training pool only, val/test pass through unchanged); Threshold Optimization by M as well, because the calibrated model—and hence its predicted probabilities—varies with the M-dependent training pool $\mathcal{S}_M$ (Eq. (7)). Per-factor level sets are listed in Table I (Section IV-B); every (R, M, D, G) combination defines one factorial cell.

where v is vehicle speed and L is the wheelbase. Lane offset evolves as:

$$\ddot{e}_{\mathrm{lane}}(t) = a_y(t) - a_{y,\mathrm{road}}(t) \tag{10}$$

Because $\dot{\delta} = d\delta/dt$ and $a_y \propto \delta$, four of the five signals (δ, $\dot{\delta}$, a_y, e_{lane}) are deterministically coupled through a single degree of freedom $\delta(t)$, with only a_x dynamically independent.

2) Data Preprocessing: Raw signals are segmented into 3-second sliding windows with 50% overlap (1.5-second step). Each window is labeled using the Karolinska Sleepiness Scale (KSS) [27]: KSS $\in \{1, \ldots, 5\}$ maps to Alert (class 0) and KSS $\in \{8, 9\}$ to Drowsy (class 1); intermediate scores (6, 7) are excluded to maximize label reliability.

For each window, the 18 statistical and spectral descriptors of Arefnezhad et al. [10] are extracted from each of the five vehicle signals (δ, $\dot{\delta}$, a_y, a_x, e_{lane}), yielding $p = 90$ candidate features for the classifier. Each signal's raw waveform and its FFT power spectrum in the $[0.5, 30]$ Hz band produce 18 descriptors: Range, StdDev, Energy, Zero-Crossing Rate, Quartiles (Q25, Median, Q75), Skewness, Kurtosis, Katz Fractal Dimension, Shannon Entropy, Frequency Variance, Spectral Entropy, Spectral Flux, Frequency Center of Gravity, Dominant Frequency, Average PSD, and Sample Entropy. The processed-data CSV additionally contains $5 \times 3 = 15$ smoothed-trend / standard-deviation / prediction-error descriptors and $5 \times 8 = 40$ Geronimo–Hardin–Massopust (GHM) multiwavelet packet decomposition energies (3-level decomposition, 8 sub-bands per signal); these 55 additional features are not loaded by the RF classifier but are used downstream for the inter-subject distance computation that defines factor D (Section IV-B1), giving the distance space its $90+15+40 = 145$ dimensions. To reduce dimensionality and mitigate overfitting, the 90 classifier-candidate features are ranked by random forest feature importance [28] (200 trees, balanced class weights) and the top $k = 10$ are retained for each training configuration, balancing discriminative power against overfitting given the per-cell sample sizes.

3) Classifier and Hyperparameter Optimization: The classifier $f\colon \mathbb{R}^p \to \{0, 1\}$ introduced in Section III is instantiated as a random forest (RF) [28] and trained on the rebalanced pool $\tilde{\mathcal{D}}$ defined above. Its hyperparameter vector $\boldsymbol{\theta}$ is tuned independently per factorial cell via Optuna [29] with 100 Tree-structured Parzen Estimator (TPE) trials and 3-fold stratified cross-validation:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}\in\Theta} \; \mathrm{F2}_{\mathrm{CV}}(\boldsymbol{\theta};\, \tilde{\mathcal{D}}) \tag{11}$$

where Θ is the search space defined in Table III. After Optuna selection and Platt-scaling calibration (described below), the classification threshold τ is swept over the set of unique calibrated posterior probabilities derived from the precision–recall curve on the validation set, retaining the value that maximizes F2:

$$\hat{\tau} = \arg\max_{\tau\in[0,1]} \; \mathrm{F2}\Big(\big\{(\hat{p}_w^{\mathrm{cal}},\, y_w) : (\mathbf{x}_w, y_w) \in \textstyle\bigcup_{s\in\mathcal{G}_{\mathrm{eval}}} \mathcal{D}_s^{\mathrm{val}}\big\},\; \tau\Big) \tag{12}$$

where $\hat{p}_w^{\mathrm{cal}}$ is the calibrated posterior probability (Platt scaling, described below). Calibration precedes threshold optimization: the base RF is calibrated via Platt scaling [30] using CalibratedClassifierCV with a fresh 5-fold stratified partition (random_state=42) of the train+validation pool, independent of the 3-fold HPO partition—which uses train only, so the validation split never enters HPO. Within each calibration fold, the base classifier is refit on 4/5 of the combined data and a sigmoid is fit on the held-out 1/5; the final calibrated probability $\hat{p}_w^{\mathrm{cal}}$ is the mean across the five fold-specific calibrators. During HPO, Optuna co-explores class_weight $\in$ {balanced, balanced_subsample} jointly with the seven other hyperparameters in Table III, so that the CV objective $\mathrm{F2}_{\mathrm{CV}}(\boldsymbol{\theta};\, \tilde{\mathcal{D}})$ is evaluated under standard imbalance-aware tree construction. For the final evaluation, the seven non-class_weight hyperparameters from $\hat{\boldsymbol{\theta}}$ are retained, but class_weight is overridden to a fixed {0:1.0, 1:3.0} across all factorial cells. This override deliberately discards the HPO-selected class_weight so that minority-recall emphasis is uniform across the factorial design, decoupling the class-weight axis from the rebalancing factor R (otherwise an Optuna selection of, e.g., balanced versus

TABLE III
Random Forest Hyperparameter Search Space (Optuna, 100 TPE Trials)

Parameter	Range	Type
n_estimators	100–1,500	int
max_depth	{5, 10, 20, 30, 50, 100, None}	cat
min_samples_split	10–100	int
min_samples_leaf	5–50	int
max_features	$\{\sqrt{k}, \log_2 k$, 0.05, 0.1, 0.2, 0.3, 0.5}	cat
min_wt_frac_leaf	0.0–0.1	float
max_samples	{None, 0.5, 0.7, 0.9}	cat
class_weight†	{balanced, balanced_subsample}	cat

† Searched during HPO but overridden to {0:1.0, 1:3.0} in the final evaluation (Section IV); the selected value is discarded and only the other seven parameters are carried forward.

balanced_subsample per factorial cell would itself become a confounding factor). Each observation in the factorial experiment is therefore:

$$Y_{R,M,D,G} = \mathrm{Metric}\Big(\big\{(\hat{y}_w,\, y_w) : (\mathbf{x}_w, y_w) \in \mathcal{D}_{\text{test}}\big\}\Big),$$
$$\hat{y}_w = \mathbb{1}\big[\hat{p}_w^{\text{cal}} \geq \hat{\tau}\big] \tag{13}$$

where $\mathcal{D}_{\text{test}} = \bigcup_{s \in \mathcal{G}_{\text{eval}}} \mathcal{D}_s^{\text{te}}$ and $\mathrm{Metric} \in \{\mathrm{F2}, \mathrm{AUROC}, \mathrm{AUPRC}\}$.

Computational cost. Each of the 1,512 training runs (126 cells × 12 seeds) involves 100 Optuna trials with 3-fold CV, totalling $\approx 4.5 \times 10^5$ model fits. Individual HPC jobs required 4 CPU cores, 8–10 GB RAM, and 6–8 hours wall time; the entire experiment consumed ≈ 10,000 core-hours on the university cluster.

Inference latency. A trained RF with ≤ 1,500 trees and $k = 10$ input features produces a prediction in < 1 ms on a single CPU core, well within the 3-second window update interval.

4) Evaluation Metrics: The observation $Y_{R,M,D,G}$ in Eq. (13) is computed with three complementary metrics chosen to capture different aspects of imbalanced classification:

F2-score is the threshold-dependent, recall-weighted harmonic mean of precision (P) and recall (Rec):

$$\mathrm{F2} = \frac{5\,P\,\mathrm{Rec}}{4\,P + \mathrm{Rec}} \tag{14}$$

The $\beta = 2$ weighting penalizes missed drowsy episodes more heavily than false alarms, reflecting the safety-critical nature of DDD. F2 is used as the primary optimization target in Eqs. (11) and (12).

AUROC (area under the ROC curve) summarizes threshold-free discrimination ability over the full operating range. It complements F2 by assessing whether the model ranks drowsy windows above alert ones regardless of threshold choice.

AUPRC (area under the precision–recall curve) [31] provides a threshold-free measure that is more informative than AUROC under severe class imbalance, because it does not credit correct majority-class predictions.

Together, these three metrics form a minimal sufficient set: one threshold-dependent recall-focused metric (F2), one threshold-free global metric (AUROC), and one threshold-free imbalance-aware metric (AUPRC).

D. Analysis Methods

Three analysis methods are applied to the 1,512 evaluation results: an intuitive OFAT scan, a Sobol–Hoeffding variance decomposition, and a statistical framework for effect sizes and null-claim support.

1) One-Factor-at-a-Time (OFAT) Analysis: Before the quantitative decomposition, we perform a one-factor-at-a-time (OFAT) analysis [32] to build intuition. For each target factor, all other factors are fixed at every possible combination, and the mean performance is plotted across the target factor's levels. Each resulting line represents one fixed condition (averaged over 12 seeds), producing a "spaghetti plot" that reveals: (i) whether the factor's effect is consistent across conditions, (ii) how much the estimate varies depending on which conditions are fixed, and (iii) the magnitude of the factor's effect relative to the spread across conditions.

2) Variance-Based Sensitivity Analysis (Sobol Indices): To quantify the relative importance of each factor and their interactions, we adopt the Sobol–Hoeffding functional ANOVA decomposition [32], [33]. Let the response surface $Y = g(X_1, \ldots, X_k)$ map factor configurations to performance, where $Y \equiv Y_{R,M,D,G}$ (Eq. (13)) and $X_i \in \{R, M, D, G\}$. The total variance admits a unique orthogonal decomposition:

$$V(Y) = \sum_{i=1}^{k} V_i + \sum_{i<j} V_{ij} + \cdots + V_{1,2,\ldots,k} \tag{15}$$

where $V_i = \mathrm{Var}_{X_i}\big[\mathbb{E}_{X_{\sim i}}(Y \mid X_i)\big]$ is the variance of the conditional expectation over factor i alone, and V_{ij} captures the joint effect of (X_i, X_j) not explained by their individual contributions. The first-order Sobol index normalizes each contribution:

$$S_i = \frac{V_i}{V(Y)} \tag{16}$$

and the total-order index aggregates the main effect with all interactions involving factor i:

$$S_{Ti} = 1 - \frac{V_{\sim i}}{V(Y)} \tag{17}$$

where $V_{\sim i} = \mathrm{Var}_{X_{\sim i}}\big[\mathbb{E}_{X_i}(Y \mid X_{\sim i})\big]$ is the variance explained by all factors except i. The difference $S_{Ti} - S_i$ quantifies the fraction of variance attributable to interactions involving factor i. Because our design is balanced, all sum-of-squares terms are computed exactly via marginal means without Monte Carlo sampling. Confidence intervals (95%) are obtained by percentile bootstrap ($B =$ 2,000) resampling over the 12 seeds.

3) Statistical Framework: Effect sizes between factor levels are measured by Cliff's δ [34] (< 0.147 negligible, < 0.33 small, < 0.474 medium, ≥ 0.474 large). Claims of factor irrelevance (e.g., distance metric equivalence) are supplemented with Bayes factors via BIC approximation [35], interpreted on the Jeffreys scale [36].

Global validation. A permutation test ($B =$ 10,000) validates the overall rebalancing effect. Seed convergence is assessed by σ_{rank}, the standard deviation of each strategy's rank position across random k-seed subsets averaged over all $\binom{12}{k}$ draws; $\sigma_{\text{rank}} \to 0$ indicates that rankings are determined by strategy identity rather than seed selection.

Per-feature inter-subject identifiability. For each of the 145 features in the distance-computation pool of Section IV-C2 (90 Arefnezhad descriptors + 15 smooth/std/pred-error + 40 wavelet packet sub-bands), we report a per-feature one-way intraclass correlation, $\text{ICC}_j = \hat{\sigma}^2_{\text{subject},j} / (\hat{\sigma}^2_{\text{subject},j} + \hat{\sigma}^2_{\text{window|subject},j})$, where $\hat{\sigma}^2_{\text{subject},j}$ is estimated as the variance across the 87 subject-means of feature j and $\hat{\sigma}^2_{\text{window|subject},j}$ as the average within-subject variance of the same feature, following the ICC(1,1) moment estimator of Shrout and Fleiss [37]. The reported overall value $\text{ICC} = (1/p)\sum_j \text{ICC}_j$ summarizes how distinguishable subjects are in the feature space used for inter-subject distance computation; this quantity is referenced once in Section VI-B.

V. Experimental Results

A. Intuitive Factor Analysis via OFAT

Before the quantitative variance decomposition, we screen each factor's effect using OFAT spaghetti plots (Fig. 3). In each panel, every thin line represents one fixed combination of the other three factors (averaged over 12 seeds); the bold line shows the grand OFAT mean and the gray band indicates ±1 SD. Columns correspond to the four experimental factors; rows show F2-score, AUROC, and AUPRC.

The two left columns show the dominant factors. Rebalancing (R): SMOTE and SW-SMOTE variants consistently outperform Baseline and RUS, but the inter-condition spread ($\Delta = 0.08$–0.55 for F2) reveals strong mode-dependence—cross-domain conditions cluster near floor performance while within/mixed conditions fan upward, suggesting an $R \times M$ interaction. Mode (M): every condition rises monotonically from Cross to Within/Mixed, with a fan-out at Within/Mixed that mirrors the rebalancing effect.

The two right columns show the negligible factors. Distance (D): the OFAT mean is flat across MMD, DTW, and Wasserstein, and condition lines remain nearly parallel. Membership (G): the mean is near-flat, and individual lines diverge in both directions, averaging to near zero.

B. Quantitative Factor Decomposition via Sobol Indices

The Sobol–Hoeffding decomposition (Fig. 4) quantifies each factor's contribution to total variance. The permutation test confirms a significant global rebalancing effect for all three metrics ($p < 0.001$).

Mode contributes the largest main effect ($S_M = 0.31$–0.51), followed by Rebalancing ($S_R = 0.24$–0.29). Both factors participate in a substantial $R \times M$ interaction (21.2% of F2-score variance, 13.8% of AUROC variance). The total-order indices show that Mode accounts for $S_{TM} = 0.48$–0.66 and Rebalancing for $S_{TR} = 0.39$–0.46 of total variance.

In contrast, Distance ($S_{TD} < 0.015$) and Membership ($S_{TG} \leq 0.031$) are negligible even when interactions are included. Residual variance (seed-to-seed variation), denoted S_ε, accounts for 7.9%–21.9% of total variance.

Defining the systematic (non-residual) variance fraction, where S_ε is the residual (seed-level) variance share:

$$\frac{S_M + S_R + S_{R\times M}}{1 - S_\varepsilon} = \frac{0.8224}{0.8433} = 97.5\% \text{ (F2)} \tag{18}$$

where S_M=0.3675, S_R=0.2425, $S_{R\times M}$=0.2124, and S_ε=0.1567. Analogous ratios are 95.8% (AUROC) and 96.1% (AUPRC), confirming that M, R, and their interaction account for $> 95\%$ of systematic variance across all three metrics. The interaction concentration ratio $C^{(i)}_{R\times M} = S_{R\times M}/(S_{Ti} - S_i) = 87$–$96\%$ indicates that the $R \times M$ pair absorbs nearly all interaction variance.

Bayes factors further confirm distance metric equivalence, providing "very strong" to "extreme" evidence for H_0 on the Jeffreys scale [36] ($BF_{01} = 71$–767).

C. $R \times M$ Interaction Analysis

The Sobol decomposition identified a single dominant interaction ($R \times M$). Table IV and Fig. 5 characterize its structure.

Within-domain and Mixed are statistically equivalent in the pooled comparison ($|\delta_{\text{F2}}| = 0.04$, $|\delta_{\text{AUROC}}| = 0.05$, $|\delta_{\text{AUPRC}}| = 0.03$), while Cross-domain collapses to near-chance levels ($|\delta| > 0.83$ vs. Within for F2; $|\delta| > 0.94$ for AUROC and AUPRC). The best strategy also reverses across modes: RUS r=0.1 ranks first in Cross-domain, whereas SW-SMOTE r=0.1 ranks first in Within and Mixed (Fig. 5). The Spearman rank correlation of strategy rankings between Cross and Within modes ranges from $\rho_{\text{cross,within}} = -0.74$ to -0.89 across metrics, confirming systematic reversal. SW-SMOTE r=0.1 with within-domain training achieves the highest threshold-independent performance (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558); Mixed-mode SW-SMOTE r=0.1 attains a slightly higher F2 (0.587) at a more recall-heavy operating point but a lower AUROC (0.879).

D. Robustness Validation

Ranking stability (σ_{rank}) decreases monotonically with the number of seeds, reaching 0 (F2, AUPRC) or 0.147 (AUROC) by $k = 11$ of 12 seeds (Fig. 6), confirming sufficient seed count.

VI. Discussion

A. Answer to RQ1: Relative Contribution of Each Factor

The Sobol decomposition reveals a concentrated variance structure: M, R, and $R \times M$ jointly explain $> 95\%$ of systematic variance (Eq. (18)), while D and G together contribute $< 5\%$.

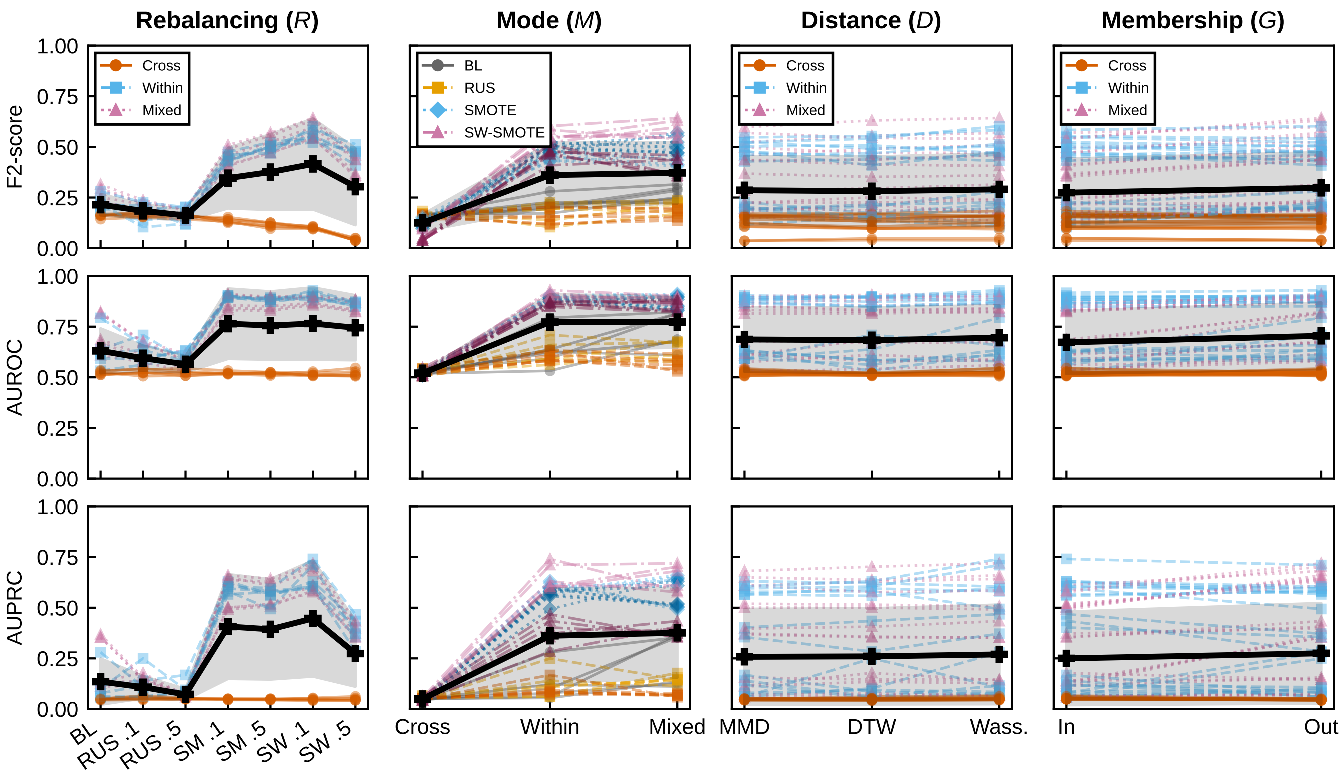

Fig. 3. OFAT spaghetti plots for all four experimental factors (columns: R, M, D, G) and three metrics (rows: F2-score, AUROC, AUPRC). Each thin line represents one fixed combination of the other three factors averaged over 12 seeds. Columns R, D, G encode lines by mode: Cross (vermilion, solid line, circle marker), Within (sky blue, dashed line, square marker), Mixed (reddish purple, dotted line, triangle marker). Column M encodes lines by rebalancing family: BL (grey, solid, circle), RUS (orange, dashed, square), SMOTE (sky blue, dotted, diamond), SW-SMOTE (purple, dash-dot, triangle); ratio variants share the same line style within each family. The bold black line (plus marker) shows the OFAT grand mean; the grey band indicates ±1 SD. Left two columns show dominant factors with large spread; right two columns show flat trajectories confirming negligible effects. The mode-based encoding in the R, D, G columns is chosen specifically to make the dominant $R \times M$ interaction visible (Cross-domain lines stay flat near chance, Within/Mixed lines fan out under SMOTE-family rebalancing); a rebalancing-family encoding is used instead in the M column because mode is the X-axis there.

TABLE IV
Mean Performance by Rebalancing Strategy, Disaggregated by Training Mode. Cross-domain results (all F2 < 0.17) are shown for completeness; the $R \times M$ interaction analysis focuses on Within and Mixed modes where strategy differentiation is meaningful.

Strategy	F2 (Cross)	F2-score Within	F2-score Mixed	AUROC Within	AUROC Mixed	AUPRC Within	AUPRC Mixed
Baseline	0.160	0.215	0.268	0.633	0.737	0.116	0.241
RUS $r = 0.1$	0.165	0.178	0.208	0.625	0.632	0.122	0.147
RUS $r = 0.5$	0.156	0.159	0.170	0.603	0.568	0.097	0.068
SMOTE $r = 0.1$	0.138	0.448	0.452	0.898	0.875	0.600	0.573
SMOTE $r = 0.5$	0.114	0.499	0.514	0.882	0.865	0.562	0.570
SW-SMOTE $r = 0.1$	0.101	0.558	**0.587**	**0.903**	0.879	**0.648**	0.646
SW-SMOTE $r = 0.5$	0.042	0.468	0.402	0.862	0.852	0.386	0.387

To understand why training mode dominates, define the subject overlap ratio between the training pool $\mathcal{S}_M$ (Eq. (7)) and the evaluation group $\mathcal{G}_{\text{eval}}$ (Eq. (6)):

$$\omega_M = \frac{|\mathcal{S}_M \cap \mathcal{G}_{\text{eval}}|}{|\mathcal{G}_{\text{eval}}|} \tag{19}$$

By construction, $\omega_{\text{Cross}} = 0$ (training excludes evaluation subjects), while $\omega_{\text{Within}} = \omega_{\text{Mixed}} = 1$ (evaluation subjects are included). This binary switch between $\omega = 0$ and $\omega = 1$ determines whether target-subject patterns are available during training, which in turn governs the performance bottleneck and the effectiveness of rebalancing (Section VI-B).

In contrast, distance metric and domain membership are negligible ($S_{TD} < 0.015$, $S_{TG} \leq 0.031$; $BF_{01} = 71$–767). Three fundamentally different metrics—MMD (distributional), DTW (temporal), and Wasserstein (geometric)—produce equivalent downstream performance despite generating different subject rankings (Table V; DTW vs. MMD: $\rho = 0.12$, $p = 0.29$).

This metric invariance stems from three factors: the bicycle model (Eqs. (9)–(10)) compresses the effective di-

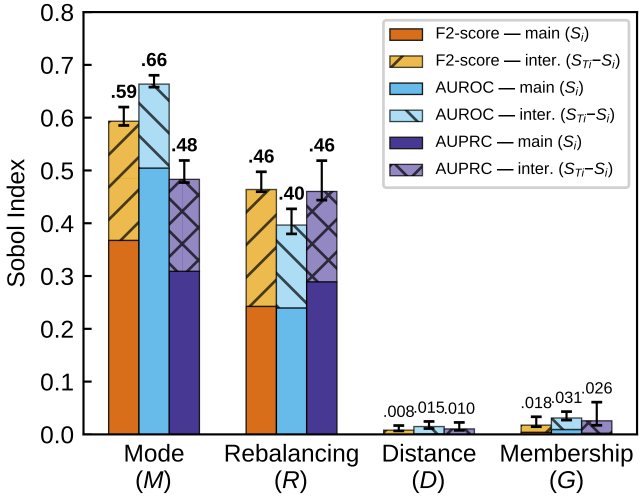

Fig. 4. Sobol sensitivity indices for the four factors. Solid bars: first-order indices (S_i); hatched bars: interaction contributions ($S_{Ti} - S_i$). Error bars: 95% bootstrap CIs on S_{Ti}. Mode and Rebalancing dominate; Distance and Membership are negligible.

TABLE V
Pairwise Rank Correlation and Domain Partition Overlap Between Distance Metrics. Jaccard index is computed on the 44-subject in-domain group; the switcher count is the number of subjects reassigned between in- and out-domain.

Pair	Spearman ρ	Jaccard	Switchers
MMD vs. W_1	0.757	0.544	26 (29.9%)
MMD vs. DTW	0.116 (n.s.)	0.443	34 (39.1%)
W_1 vs. DTW	0.444	0.725	14 (16.1%)

mensionality of the 145-dimensional distance-computation feature space (Section IV-C2) to $p_{\text{eff}} = 49$ (95% PCA variance, with the first principal component alone explaining 20.4%), low per-feature inter-subject identifiability (mean ICC = 0.111 across the same 145 features; see Section IV-D) makes partition relabelling inconsequential despite up to 39% of subjects switching domain (Table V), and the dominant rebalancing factor absorbs residual grouping differences ($S_{TR}/S_{TD} = 26$–$31\times$; $BF_{01} > 70$).

B. Answer to RQ2: Interaction and Optimal Factor Combination

The $R \times M$ interaction ($S_{R\times M} = 0.212$ for F2, concentrated at 87–96% of each dominant factor's interaction variance) confirms that the contributing factors do not act independently. This interaction manifests as a mode-dependent strategy reversal: the identity of the best rebalancing strategy changes across training modes (Table IV; Fig. 5). A practitioner who selects a rebalancing strategy based on cross-domain results would deploy the worst strategy for within-domain operation, and vice versa.

The mechanistic origin is a mode-dependent performance bottleneck. The mode-dependent rebalancing gain, averaged over D and G, is:

$$\Delta Y_R^{(M)} = \mathbb{E}_{D,G}[Y_{R,M,D,G} - Y_{\text{BL},M,D,G}] \tag{20}$$

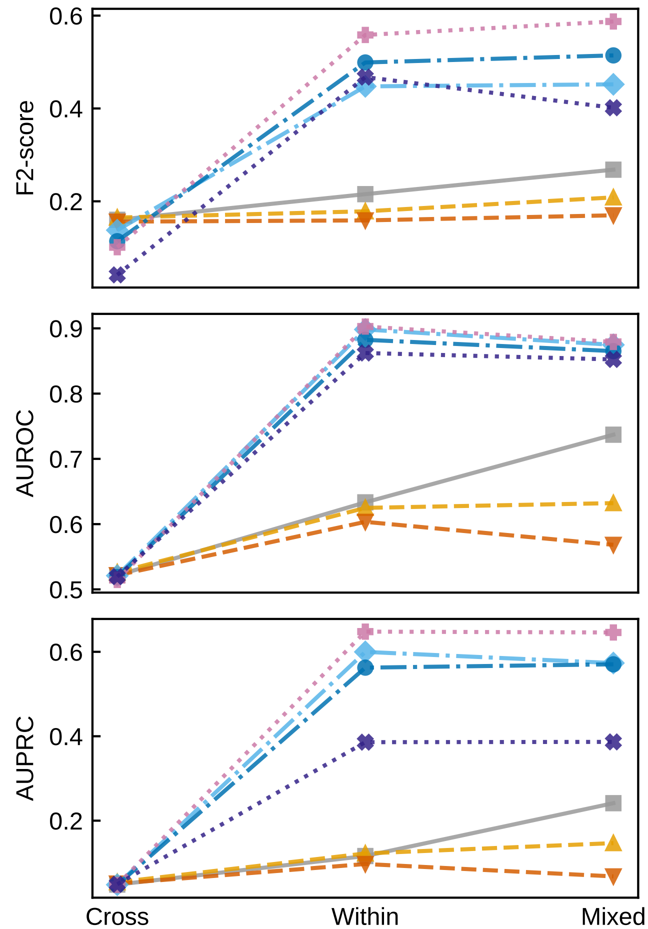

Fig. 5. Mode-dependent strategy performance across training modes. Line styles encode rebalancing families: solid = Baseline (grey), dashed = RUS (orange/vermilion for r=0.1/0.5), dash-dotted = SMOTE (light blue/blue for r=0.1/0.5), dotted = SW-SMOTE (pink/purple for r=0.1/0.5). Markers further distinguish strategies: square, up/down triangle, diamond, circle, plus, and cross. In Cross-domain, all strategies perform poorly; in Within/Mixed, SMOTE-based strategies sharply improve while RUS variants remain low—illustrating the $R \times M$ interaction.

For the best strategy (SW-SMOTE r=0.1), the empirical values are:

$$\Delta Y_{\text{SW-SM.}}^{(M)} = \begin{cases} +0.343\ (+159\%) & M = \text{Within}\ (\omega = 1) \\ +0.319\ (+119\%) & M = \text{Mixed}\ \ (\omega = 1) \\ -0.059\ (-37\%) & M = \text{Cross}\ \ \ (\omega = 0) \end{cases} \tag{21}$$

When $\omega_M = 1$, the performance bottleneck is class imbalance ($\rho \ll 1$), and oversampling is effective. When $\omega_M = 0$, the bottleneck shifts to distributional mismatch ($P_{\text{pool}}(\mathbf{x}) \neq P_{\text{test}}(\mathbf{x})$), and all strategies collapse to near-chance levels (F2 = 0.04–0.17). This sign reversal of $\Delta Y_R^{(M)}$ (Eq. (21)) is the mechanistic origin of the $R \times M$ interaction. The $G \times M$ interaction is negligible ($S_{G\times M} = 0.006$–0.009), an order of magnitude below $S_{R\times M}$. The optimal factor combination is therefore SW-SMOTE r=0.1 with $\omega_M = 1$; within-domain training maximizes threshold-independent discrimination (AUROC =

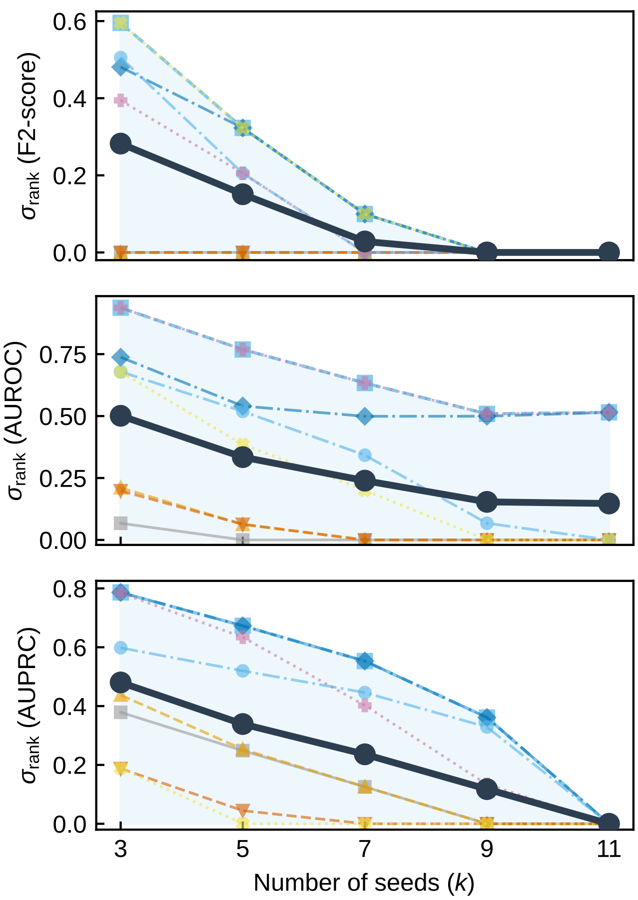

Fig. 6. Ranking stability (σ_{rank}) as a function of seed subset size k. Thick solid line with circles: mean σ_{rank} across all seven strategies; dashed line with squares and shaded band: maximum σ_{rank}. Thin lines show per-strategy convergence (line style encodes family): solid with square = Baseline; dashed with up/down triangles = RUS r=0.1/0.5; dash-dotted with diamond/circle = SMOTE r=0.1/0.5; dotted with plus/cross = SW-SMOTE r=0.1/0.5. Monotonic convergence confirms that $n = 12$ seeds is sufficient.

0.903, AUPRC = 0.648; within-domain F2 = 0.558), while Mixed mode trades 0.024 AUROC for a higher recall-leaning F2 (0.587).

C. Practical Implications

The Sobol total-order indices establish a strict factor priority ordering for practitioners deploying vehicle-dynamics-based DDD systems:

$$\begin{aligned} S_{TM}\,(0.48\text{–}0.66) \;&>\; S_{TR}\,(0.39\text{–}0.46) \\ &\gg\; S_{TG}\,(\leq 0.031) \;\approx\; S_{TD}\,(<0.015) \end{aligned} \tag{22}$$

This ordering, combined with the subject overlap ratio ω_M (Eq. (19)), yields the following decision rules:

1) Training mode (highest priority): Use within-domain or mixed training whenever target-subject data is available ($\omega_M = 1$).
2) Class rebalancing (second priority): The optimal rebalancing strategy is mode-dependent (Eq. (21)). When $\omega_M = 1$, apply SW-SMOTE at $r = 0.1$:

$$R^*(\omega_M) = \begin{cases} \text{SW-SMOTE, } r{=}0.1 & \omega_M{=}1 \\ \text{any (none helps)} & \omega_M{=}0 \end{cases} \tag{23}$$

where $\Delta Y = +159\%$ (F2) for $\omega_M = 1$ and $\Delta Y \leq 0$ for $\omega_M = 0$.
3) Distance metric and domain membership (negligible): Use any convenient metric for domain grouping. The choice has no measurable impact on classification performance ($BF_{01} > 70$).
4) Bridging the cross-domain gap ($\omega_M = 0$). Since no tested rebalancing strategy improves cross-domain performance (Eq. (21)), closing this gap requires reducing the distributional mismatch $P_{\text{pool}}(\mathbf{x}) \neq P_{\text{test}}(\mathbf{x})$ directly. Promising directions include unsupervised domain adaptation [6], [7], few-shot personalization [38], and continual learning. Each can be introduced as an additional factor level within the same Sobol decomposition pipeline, yielding directly comparable variance shares.

D. Comparison with Prior Work

Our finding that class imbalance handling is the dominant design factor aligns with Taylor et al. [39], who showed that data-related hyperparameters dominate architectural choices in deep learning. However, our study is the first to demonstrate this in the DDD context and to reveal the interaction with training mode that causes mode-dependent strategy ranking change. Prior DDD studies that compare rebalancing methods under a single training mode [18], [40] risk selecting a suboptimal strategy for deployment.

The best within-domain performance (F2 = 0.558, AUROC = 0.903) is competitive with recent vehicle-dynamics-based methods. Table VI positions this work relative to prior DDD studies. Direct performance comparison is limited because prior studies use different datasets, modalities, and evaluation protocols; the primary contribution here is the factorial decomposition framework, not absolute classification performance.

The distance metric irrelevance contrasts with the implicit assumption in domain adaptation literature [6], [38] that domain definition substantially impacts downstream performance; the three-layer mechanism identified in Section VI-A explains why this assumption does not hold for vehicle-dynamics features.

E. Limitations

1) Single classifier family. The factorial experiment uses RF [28]. The Sobol–Hoeffding framework is classifier-agnostic and can be extended to other learners (e.g., LSTM, SVM) without methodological modification.

TABLE VI
Comparison with Prior DDD Studies. "Factors" Indicates the Number of Design Factors Systematically Varied; "Interaction" Indicates Whether Inter-Factor Interactions Were Quantified. Most prior studies report Accuracy and/or F1 instead of F2/AUROC/AUPRC; the "Other reported" column lists the headline metric(s) actually given in the source paper. NR = not reported.

Study	Year	Classif.	Modal.	Subj.	Fact.	Inter.	F2	AUROC	AUPRC	Other reported
Arefnezhad et al. [10]	2019	SVM	Vehicle	39	0	No	NR	0.97†	NR	Acc=98.1%
Zhu et al. [15]	2025	RF	Vehicle	25	0	No	NR	NR	NR	Acc/P/R = 93.7/93.7/93.9%
Baccour et al. [11]	2022	SVM	Vehicle	70	0	No	NR	NR	NR	Bal. Acc=87.1%
Sun et al. [12]	2021	Fisher	Vehicle	35	0	No	NR	NR	NR	Acc=85.3%, F1=79.6%
Yadawadkar et al. [18]	2018	NB	Vehicle	—	1 (imbalance)	No	NR	NR	NR	F1=94% (NB+SMOTE)
Schwarz et al. [19]	2019	RF	Vehicle	20	0	No	NR	NR	NR	5-fold Acc=0.94–0.95
He et al. [40]	2024	GARMFCN	EEG+EOG	23	1 (imbalance)	No	NR	0.87–0.96§	NR	Acc=84.7%, w-F1=0.85
Kim et al. [38]	2025	PDMix	EEG	27	1 (domain)	No	NR	0.738‡	NR	F1=70.3%
Ours	2026	RF	Vehicle	87	4 (R,M,D,G)	$R\times M$ Sobol	0.558	0.903	0.648	—

† Evaluation protocol not specified in source; based on the reported sample counts (35,079 test samples drawn from 39 drivers' 20 h of pooled data), the most plausible default is a window-level random split (effectively within-subject), which is not directly comparable to our per-subject 70/15/15 chronological split.

‡ Cross-subject evaluation (PDMix is a domain-generalization method evaluated on subject-disjoint splits); directly comparable to our cross-domain (ω_M=0) results, not the within-domain AUROC = 0.903 reported in the same row.

§ Per-class one-vs-rest AUC for a 3-class (awake/tired/drowsy) classifier on EEG+EOG; not directly comparable to our binary AUROC. The reported per-class values are 0.87 (awake), 0.88 (tired), 0.96 (drowsy).

2) Deterministic data split. The train/test partition is deterministic; random seeds vary only model initialisation and resampling.
3) Simulator data. All data come from a laboratory driving simulator. On-road conditions introduce additional variability that may alter the absolute variance shares; validation on naturalistic datasets (e.g., SHRP 2) is needed.
4) Sample size per cell. With $n = 12$, only large effects ($|\delta| > 0.53$) are detectable after Bonferroni correction.
5) Feature selection scope. Only the top $k = 10$ features from $p = 90$ Arefnezhad-style candidates are used in the classifier (the additional 55 smooth/std/pred-error and wavelet packet features computed in preprocessing serve only the distance computation); sensitivity to k and to the unified 145-dimensional feature set as a single classifier pool remains to be tested.
6) Binary labeling threshold. KSS $\leq$ 5 vs. KSS $\geq$ 8 excludes intermediate drowsiness (KSS 6–7), which may limit applicability to graded drowsiness detection scenarios.
7) Calibration–threshold coupling. When $M \in \{\text{Within}, \text{Mixed}\}$, the calibration pool $\mathcal{S}_M$ includes $\mathcal{G}_{\text{eval}}$, so $\mathcal{G}_{\text{eval}}$'s validation windows serve both as part of the 5-fold calibration data and as the support for the F2-optimal threshold $\hat{\tau}$; $\hat{\tau}$ is therefore mildly optimistic on the validation set for these two modes. When $M =$ Cross, calibration and threshold-optimization pools are drawn from disjoint subject sets, so no coupling arises. Final test metrics are unaffected in all modes because the test split is held out from HPO, calibration, and threshold selection.

VII. Conclusion

This study quantifies the relative importance of four design factors in vehicle-dynamics-based DDD through a factorial experiment with Sobol–Hoeffding variance decomposition over 1,512 random forest runs from 87 drivers. Training mode (S_{TM} = 0.48–0.66) and rebalancing (S_{TR} = 0.39–0.46), including their interaction ($S_{R\times M} = 0.21$), account for $> 95\%$ of systematic variance, whereas distance metric ($S_{TD} < 0.015$; BF_{01} = 71–767) and domain membership ($S_{TG} \leq 0.031$) are negligible. The $R\times M$ interaction reverses the optimal strategy across modes ($\rho_{\text{cross,within}} = -0.74$ to -0.89): when evaluation subjects overlap with training data ($\omega_M = 1$), class imbalance is the bottleneck and oversampling is effective; when they do not ($\omega_M = 0$), distributional mismatch dominates and no tested rebalancing strategy helps. SW-SMOTE r=0.1 with within-domain training ($\omega_M = 1$) achieves the highest threshold-independent performance (AUROC = 0.903, AUPRC = 0.648; within-domain F2 = 0.558). Practitioners should prioritize training mode and class rebalancing (Eq. (22)); domain configuration requires no optimization.

Future work includes validation with other classifiers (LSTM, SVM), on naturalistic driving data (SHRP 2), and incorporation of domain adaptation and few-shot methods as additional factor levels.

Data and Code Availability

The driving simulator dataset is publicly available at Harvard Dataverse (DOI: 10.7910/DVN/HMZ5RG) [25]. Analysis code and trained models will be released upon publication.

Acknowledgment

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